Gdańsk University of Technology
Faculty of Electronics, Telecommunications and Informatics
Department of Multimedia Systems

Research Report

AI - Features, Methods & Resources

v1.0

Report 1: Input Features, Methods and Project Resources

Project Subject: Glaucoma detection using intelligent analysis of fundus images

Team Members: Martyna Borkowska (Lead), Karolina Glaza, Mateusz Dobry, Agnieszka Pawłowska, Amila Amarasekara, Antoni Naczke

Supervisor: Prof. Andrzej Czyżewski, Ph.D., D.Sc., Eng.

Authors: Martyna Borkowska, Agnieszka Pawłowska, Karolina Glaza, Amila Amarasekara, Antoni Naczke, Mateusz Dobry

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Contents

1 Introduction

Choosing appropriate input features is a critical step in building diagnostic models for automatic glaucoma detection. Quantitative analysis of optic nerve head morphology provides key biomarkers, the most fundamental of which is the Cup-to-Disc Ratio (CDR) - the proportion between the optic cup and the whole optic disc.

2 Input Parameters

2.1 Cup-to-Disc Ratio (CDR)

The CDR comes in two main variants.

2.1.1 Vertical Cup-to-Disc Ratio (vCDR)

vCDR is the ratio of the vertical cup diameter to the vertical disc diameter.

- **High clinical sensitivity:** nerve-fibre loss in glaucoma usually starts in the polar sectors (superior and inferior), so the cup enlarges quickly along the vertical axis.
- It is also the parameter traditionally estimated by clinicians.
- **Drawback - sensitivity to anatomical variants:** in patients with irregularly shaped or tilted discs, or with large discs (macrodiscs), the vertical measurement alone can be misleading and yield a high vCDR despite no pathology.

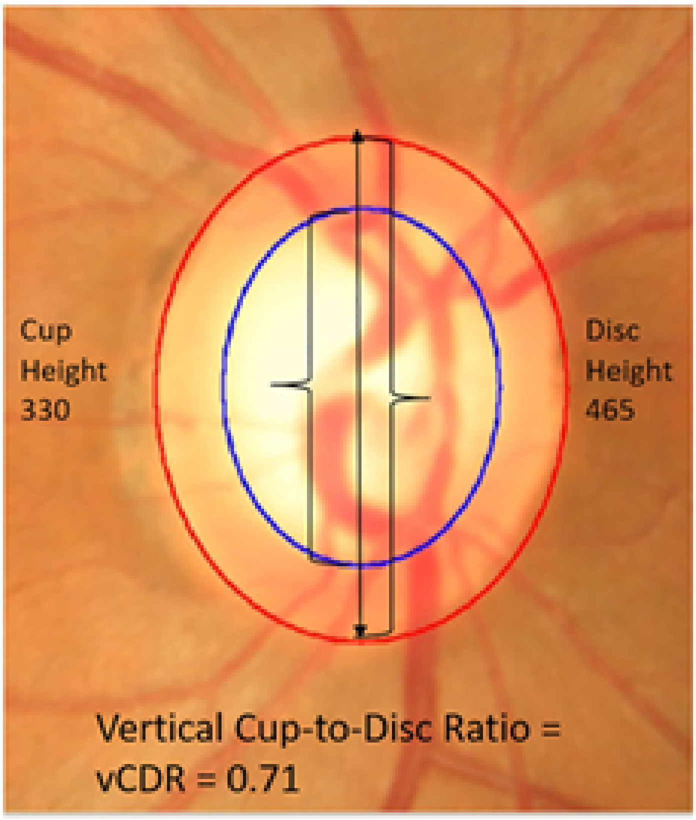

Figure 1: Illustration of the Vertical Cup-to-Disc Ratio (vCDR).

2.1.2 Area Cup-to-Disc Ratio (ACDR)

ACDR is computed as the ratio of the cup area to the total disc area.

- Much greater stability and objectivity, as it accounts for the whole cup morphology, not just one axis.
- Less sensitive to irregular disc shape, and correlates better with the total axon count.
- A "safer" parameter - for a macrodisc, where the cup is proportionally large but healthy, ACDR reflects the true state better than a potentially inflated vCDR.
- **Limitation:** it requires precise computer segmentation of both contours (disc and cup).

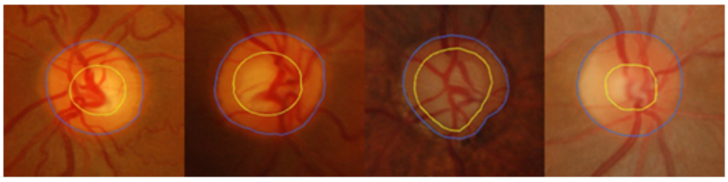

Figure 2: Examples of disc and cup segmentation for ACDR.

2.2 Neuroretinal Rim (NRR) Analysis

An alternative approach is the direct analysis of the neuroretinal rim (NRR) - the active nerve tissue.

2.2.1 Rim-to-Disc Ratio (RDR) and Rim-to-Disc Area Ratio (RADAR)

- **RDR** is the ratio of the rim area to the whole disc area - a very intuitive measure (higher is better). Note that RDR is the mathematical complement of ACDR (RDR $\approx 1 -$ ACDR), which can mean feature redundancy if both are used in a model.
- **RADAR** is the ratio of the neuroretinal rim area to the total optic disc area. It is considered a more precise indicator of rim loss than the traditional RDR because it uses area rather than just diameter, reducing the influence of disc size.
- RDR is a linear measure (ratio of the narrowest rim width to the disc diameter), whereas RADAR is an area measure - more precise and less dependent on disc shape.
- RADAR is especially useful in automatic (AI) analysis, where image segmentation allows the rim and disc boundaries to be determined accurately.

2.2.2 Disc Damage Likelihood Scale (DDLS)

The DDLS rates the likelihood of optic disc damage based on its appearance.

- Frequently used in AI models and correlates well with glaucoma progression.
- Enables a fast, objective assessment of the degree of disc damage, which can improve a model's sensitivity and specificity.

2.2.3 Sectoral NRR Analysis (ISNT Rule)

A much more advanced method is sectoral NRR analysis, based on rim thickness in four quadrants: Inferior, Superior, Nasal, and Temporal.

- A healthy disc typically follows the **ISNT rule** ($I > S > N > T$).
- Glaucoma often causes focal NRR atrophy, breaking this rule (e.g. $I < N$) - a strong sign of pathology even when global indices such as ACDR are still normal.
- This method therefore has very high diagnostic power.
- **Fundamental limitation:** it requires knowing the eye laterality - the model must know whether it analyses the right (OD) or left (OS) eye to correctly identify the nasal and temporal quadrants.

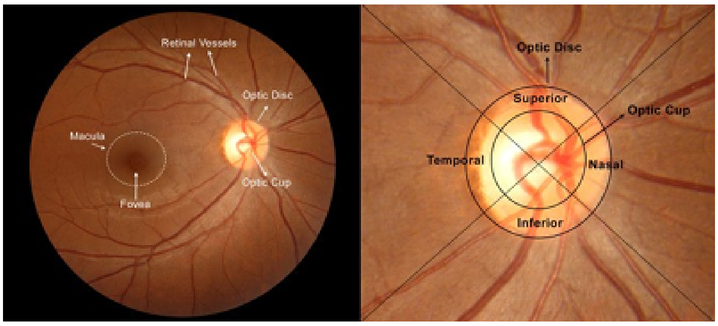

Figure 3: Fundus anatomy and the quadrants of the optic nerve head.

2.3 Additional Image Features

Additional data may improve model accuracy:

- Haemorrhages (small bleeds near the optic disc, visible on fundus photographs).
- Damage to the nerve-fibre layer.
- Inter-eye disc asymmetry.
- Changes in disc colour.

2.4 Functional Data (if available)

- Mean Deviation (MD) from the visual field test.
- Intraocular Pressure (IOP).
- Patient age.

Such data can improve model accuracy, especially in disease-progression analysis.

2.5 Summary of Features

- **ACDR** is the most stable and universal index, reflecting the disc state regardless of its size or shape.
- **vCDR** is very sensitive to early changes but less robust to anatomical variants.
- **Sectoral NRR analysis (ISNT)** detects focal changes that may be the first sign of glaucoma, though it requires precise segmentation and eye-laterality identification.
- **RADAR and DDLS** are modern indices that increase objectivity and accuracy, especially for automatic diagnosis.
- Additional image features (haemorrhages, disc asymmetry, colour changes) can substantially improve sensitivity in hard-to-recognise cases.

Figure 4: Fundus photograph with visible haemorrhages (retinal bleeding) in the upper-right quadrant.

3 Potential Methods

A wide spectrum of classical machine-learning and deep-learning methods can be applied to automatic glaucoma detection in fundus images. The most commonly used architectures and algorithms include:

- Support Vector Machines (SVM), Random Forest, k-Nearest Neighbors (k-NN), Logistic Regression, Decision Trees.
- VGG (e.g. VGG16, VGG19), ResNet (e.g. ResNet50), Xception, Inception and Inception-ResNet, DenseNet, MobileNet (V1-V3), EfficientNet (B0-B7).
- U-Net and Attention U-Net (segmentation), 3D CNN (for OCT data).
- Vision Transformers (ViT, Swin Transformer, DeiT).
- Hybrid models (CNN + SVM, CNN + RF), GAN-based methods (augmentation or reconstruction), custom CNNs.

3.1 Open Questions

- Which of the presented methods provides the best effectiveness in glaucoma detection?

4 Databases and Data

4.1 Resources

We collected 16 free databases. Access to 2 of them (including the largest) requires a request. Most are tagged for educational use only; some are not intended for commercial use. In addition:

- All databases provide labels.
- 9 include information about the camera used to take the photographs.
- 1 database has labelled photographs of both of a patient's eyes.
- 2 databases include opinions/annotations from several physicians.
- 2 databases were used in a project identical to ours and include reports.
- 1 database contains ready binary cup and disc images.
- 3 databases are composed of several other databases.

4.2 Open Questions

- What do the clinic's database photographs look like? Are they cropped? Already converted to binary images? Do they include both of a patient's eyes?
- How much do we prioritise image processing? Should we start building the model on ready binary images, or process the photographs ourselves?
- Should we request access to the largest database and obtain details about its exact content?
- Is access to both of a patient's eyes important for us, to detect genetic dependencies?

5 Camera

Resources gathered for a low-cost fundus camera prototype: a video tutorial, an article, and a preliminary cost estimate.

Item	**Approx. cost (PLN)**
Lens (link) - ~60 USD	200-250
Phone case	35
Glue	10
Large PVC reducer (1 pc.)	4
Small PVC reducers (2 pcs.)	8
PVC pipes (50 mm and 40 mm)	15
Bolt, nut and washer	5
Cardboard	2
Sandpaper	3
Insulating tape	4
Estimated minimum total	**≈350**

5.1 Open Questions

- What funding can we get? How do we apply?
- Could we later use the university's 3D printer?
- What is the goal of this camera (acquiring photographs, promotion)?

6 Rights to the Model

While searching for image databases, numerous "for educational use only" markings were encountered. Based on a consultation with Krzysztof Cwalina, Ph.D., Eng. during a lecture, it was established that a trained model would not belong exclusively to the project team, since it is a derivative of the data it was developed on. Accordingly, an analysis of European Union regulations on the rights to data and data-based models was carried out.

- **Directive (EU) 2019/790 (DSM Directive), Articles 3 and 4.** *"Member States shall provide for an exception [...] for reproductions and extractions made by research organisations and cultural heritage institutions [...] for the purposes of scientific research." "The exception [...] shall apply on condition that the use of works and other subject matter [...] has not been expressly reserved by their rightholders."* - Using data for scientific purposes is permissible, while commercial use requires the rightholder's consent.
- **Regulation (EU) 2023/2854 - Data Act, recitals 5 and 6.** *"Users [...] can access and use the data, including by sharing them with third parties of their choice [...] imposing the obligation on data holders to make data available."* - The rights to data and to models built on them depend on agreements between the parties, rather than belonging exclusively to one of them.

Because the project is ultimately intended for a private clinic - i.e. used beyond educational purposes - a preliminary decision was made to develop two separate models:

- **Model 1** - built using data marked "for educational use only", serving research and educational purposes only and letting the team gain experience.
- **Model 2** - planned for a later phase, using data provided by the clinic.

7 Summary

This report establishes the diagnostic feature set (with ACDR as the most robust index, complemented by NRR/ISNT analysis and modern indices such as RADAR and DDLS), surveys candidate ML/DL methods, inventories the available datasets, sketches a low-cost camera prototype, and clarifies the legal constraints - leading to the two-model strategy (educational vs. clinic-data model).